

gurps-ink

Typeset unofficial *GURPS* material
User guide and reference, version 0.1.0

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1 What gurps-ink does

gurps-ink is a Typst package for people who write unofficial **GURPS** material: a sourcebook, an adventure or a handout. It does three things.

Task	What gurps-ink gives you	Chapter
Legal notices	The disclaimer and the notice that the Steve Jackson Games online policy asks for, and GURPS in bold italics.	Section 3
Dice and books	Dice such as 2d-1, damage from ST, and references such as GURPS Magic p. 14.	Section 4
Characters	Stat blocks that calculate their point costs.	Section 5

It does not do these things:

- It does not copy the look of **GURPS** books. The Steve Jackson Games online policy does not let you do that (Section 3.4). gurps-ink follows the typographic rules of Steve Jackson Games, and you choose the fonts and layout.
- It does not give legal advice.
- It does not calculate the cost of advantages and disadvantages. You give these costs.
- It does not import characters from GURPS Character Sheet (GCS).

1.1 How to use this manual

Start with Section 2. Then go to the chapter for your task. Each chapter starts with a list of the questions that it answers. Each section answers one question.

In each example, the code is on the left and the result is on the right. Each example in chapters 2 to 5 is complete: copy it into a new file and it compiles. A reference example (Section 7) needs the import line first.

The examples use one story: a race between the Tortoise and the Hare. Section 6 shows the complete one-shot adventure that they come from.

1.2 Find your question

I want to ...	Go to
install gurps-ink	Section 2
add the notices that Steve Jackson Games asks for	Section 3.1
know if my book can look like a GURPS book	Section 3.4
write dice or damage	Section 4.1, Section 4.2
refer to a GURPS book or page	Section 4.3
make a stat block	Section 5.1
know which costs I must give	Section 5.2
hide point costs in a handout	Section 5.3
write a character's numbers in my text	Section 5.4
use one character in many chapters	Section 5.5
keep a character in a point budget	Section 5.6
keep a stat block in one column	Section 5.7
change the look of a stat block	Section 5.8

make a cast list	Section 5.9
understand an error	Section 5.10
see a complete adventure	Section 6
look up a function	Section 7

2 Get started

This chapter answers:

- How do I install gurps-ink?
- What does a first page look like?

2.1 Install gurps-ink

gurps-ink is not on Typst Universe yet. Until it is, install it on your computer:

1. Install [Typst](#) and [just](#).
2. Clone the [repository](#).
3. In the repository, run `just install`.
4. In your document, write `#import "@local/gurps-ink:0.1.0": *`.

When gurps-ink is on Typst Universe, you do not install it. Write `#import "@preview/gurps-ink:0.1.0": *` and Typst downloads it. This also works in the [Typst web app](#). The examples in this manual use this import.

2.2 Make your first page

Copy this example into a new file and compile it. It makes a character, shows its stat block and adds the disclaimer.

```
#import "@preview/gurps-ink:0.1.0": *

#let tortoise = character(
  name: "The Tortoise",
  ht: 13,
  basic-move: 1,
  skill("Hiking", 14, "HT/A"),
)

#stat-block(tortoise)

The Tortoise walks at Move
#level-of(tortoise, "Basic Move").

#text(size: 7pt, sjgames-disclaimer)
```

The Tortoise (14 points)

ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 13 [30].
HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 13 [0].
Basic Speed 5.75 [0]; Basic Move 1 [-20]; Dodge 8.
SM 0; DR 0; thr 1d-2; sw 1d.
Skills: Hiking-14 [4].

The Tortoise walks at Move 1.

The material presented here is my original creation, intended for use with the [GURPS](#) system from [Steve Jackson Games](#). This material is not official and is not endorsed by Steve Jackson Games.

The import line gives you all functions of gurps-ink. The next three chapters show what you can do with them.

3 Legal notices and trademarks

This chapter answers:

- Which notices must my material have? (Section 3.1)
- What if my material is a free game aid? (Section 3.2)
- How do I write “GURPS”? (Section 3.3)
- Can my book look like a GURPS book? (Section 3.4)
- Where does the text of the notices come from? (Section 3.5)

gurps-ink gives you text. It does not give legal advice. Read the [Steve Jackson Games online policy](#) before you publish. The policy is for free material. To sell material, read its section VII, “Licensed Use”.

3.1 Add the disclaimer and the notice

Section IV of the online policy, “Notices and Disclaimers”, asks for two texts on material that is not official:

- A *disclaimer*: the material is not official and Steve Jackson Games does not endorse it. Use `sjgames-disclaimer`.
- A *notice*: **GURPS** is a trademark of Steve Jackson Games. Use `sjgames-notice`.

Both texts have links to the Steve Jackson Games pages, as the policy asks. Put them on the title page or on the last page. You can make them small.

```
#import "@preview/gurps-ink:0.1.0": *  
  
#set text(size: 7pt)  
#sjgames-disclaimer  
  
#sjgames-notice
```

The material presented here is my original creation, intended for use with the **GURPS** system from [Steve Jackson Games](#). This material is not official and is not endorsed by Steve Jackson Games.

GURPS is a registered trademark of [Steve Jackson Games](#), and is copyrighted by Steve Jackson Games. All rights are reserved by Steve Jackson Games. This material is used here in accordance with the [SJ Games online policy](#).

3.2 Add a game aid notice

`sjgames-game-aid` is a notice for a free game aid, such as a character sheet or a tool. Give it the name of the author. The online policy does not give this text. Many free **GURPS** game aids use it, for example the GURPS system for Foundry VTT.

```
#import "@preview/gurps-ink:0.1.0": *  
  
#set text(size: 7pt)  
#sjgames-game-aid[Jane Doe]
```

GURPS is a trademark of [Steve Jackson Games](#), and its rules and art are copyrighted by [Steve Jackson Games](#). All rights are reserved by Steve Jackson Games. This game aid is the original creation of Jane Doe and is released for free distribution, and not for resale, under the permissions granted in the [Steve Jackson Games Online Policy](#).

3.3 Write GURPS and other trademarks

The online policy asks you to show each trademark in bold, italic or colour each time you use it. `gurps` writes **GURPS** in bold italics. `sjgames` writes Steve Jackson Games. For other trademarks, use `strong` and `emph`.

```
#import "@preview/gurps-ink:0.1.0": *  
  
This is a one-shot for #gurps.  
#sjgames publishes  
#strong(emph[Munchkin]) too.
```

This is a one-shot for **GURPS**. Steve Jackson Games publishes **Munchkin** too.

Do not put #gurps in italic text. In italic text, emph makes it upright.

3.4 Use your own look

Section II of the online policy says that if your material looks like a Steve Jackson Games product, it is “over the line”. For example, do not copy the covers of **GURPS** books.

gurps-ink does not copy the look of **GURPS** books. It sets no fonts, colours or page layout. It follows the typographic rules of Steve Jackson Games: how to write dice, book titles, page references and stat blocks. Choose your own fonts, colours and layout.

3.5 Where the text comes from

Function	Source
sjgames-disclaimer	The example disclaimer in section IV of the online policy .
sjgames-notice	The example notice in section IV of the online policy , without the sentence about art.
sjgames-game-aid	Not from the policy. The notice that many free GURPS game aids use.
gurps, sjgames	The trademark rules in the online policy .

4 Dice and book references

This chapter answers:

- How do I write dice, such as 3d or 2d-1? (Section 4.1)
- How do I write the damage for a ST? (Section 4.2)
- How do I refer to a *GURPS* book or a page of it? (Section 4.3)
- Where do these rules come from? (Section 4.4)

4.1 Write dice

Use `dice(count, modifier)`. The modifier is optional. A negative modifier has a true minus sign, not a hyphen. The result does not break across lines.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
The Hare rolls #dice(3) against  
its self-control number.  
Its kick does #dice(1, -4) cr.  
With a symbol: #dice(2, $n$).
```

The Hare rolls 3d against its self-control number. Its kick does 1d-4 cr. With a symbol: 2d+n.

4.2 Write damage from ST

`thrust(st)` and `swing(st)` give the basic damage for a ST, from the table on p. B16. They cover ST 1 to 100. Spread the result into dice with ...

```
#import "@preview/gurps-ink:0.1.0": *
```

```
At ST 6, the Hare kicks for  
#dice(..thrust(6)) cr.  
At ST 13, a sword swings for  
#dice(..swing(13)) cut.
```

At ST 6, the Hare kicks for 1d-4 cr. At ST 13, a sword swings for 2d-1 cut.

A character from `character()` has its damage in `thr` and `sw` (Section 5.4).

4.3 Refer to a GURPS book

Use `gurps-book(title, pages)`. Do not write “GURPS” in the title: the function adds it. The pages are optional. One page gets “p.”; more pages get “pp.”.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
You need the #gurps-book("Basic Set").  
See #gurps-book("Magic", 14) and  
#gurps-book("Fantasy", (12, 20)).
```

You need the *GURPS Basic Set*. See *GURPS Magic* p. 14 and *GURPS Fantasy* pp. 12, 20.

For a page of the *Basic Set*, the style of Steve Jackson Games is “p. B348”. Write it as text: p.~B348. The ~ keeps “p.” and the number on one line.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
Roll a Quick Contest (p.~B348).
```

Roll a Quick Contest (p. B348).

4.4 Where the rules come from

The dice notation and the bold italic book titles follow the [Steve Jackson Games Authors’ Guidelines: Editorial Style](#). The damage table is on p. B16 of the *GURPS Basic Set*.

5 Characters and stat blocks

This chapter answers:

- How do I make a character and show its stat block? (Section 5.1)
- Which point costs does gurps-ink calculate? (Section 5.2)
- How do I hide the point costs? (Section 5.3)
- How do I write a character's numbers in my text? (Section 5.4)
- How do I use one character in many chapters? (Section 5.5)
- How do I keep a character in a point budget? (Section 5.6)
- How do I stop a column break from splitting a stat block? (Section 5.7)
- How do I change the look of a stat block? (Section 5.8)
- How do I make a cast list or another layout? (Section 5.9)
- What does an error message mean? (Section 5.10)

A character has two steps. `character()` makes a dictionary and calculates the costs. `stat-block()` shows the dictionary. Because the two steps are separate, you can make a character once and use it in many places.

5.1 Make a character and show it

Give the attributes as named arguments. Give only the attributes that are not the default. Give the traits, skills, spells and attacks as positional arguments, in any order.

```
#import "@preview/gurps-ink:0.1.0": *

#let hare = character(
  name: "The Hare",
  st: 6, dx: 13, ht: 11,
  sm: -2,
  basic-move: 9,
  disadvantage(
    "Overconfidence (12)",
    points: -5,
  ),
  quirk("Naps after lunch"),
  skill("Running", 16, "HT/A"),
  melee-attack("Kick", 13,
    [#dice(1, -4) cr], reach: "C"),
)
#stat-block(hare)
```

The Hare (59 points)

ST 6 [-40]; DX 13 [60]; IQ 10 [0]; HT 11 [10].
HP 6 [0]; Will 10 [0]; Per 10 [0]; FP 11 [0].
Basic Speed 6.00 [0]; Basic Move 9 [15]; Dodge 9.
SM -2; DR 0; thr 1d-4; sw 1d-3.

Disadvantages: Overconfidence (12) [-5].

Quirks: Naps after lunch [-1].

Skills: Running-16 [20].

Attacks:

Kick (13): 1d-4 cr. Reach C.

The functions for traits are advantage, disadvantage, perk and quirk. For skills and spells, they are `skill` and `spell`. For attacks, they are `melee-attack` and `ranged-attack`. The reference (Section 7.3) shows all their arguments.

5.2 Know which costs are calculated

<code>character()</code> calculates	You give
The cost of each attribute and secondary characteristic	The cost of each advantage and disadvantage, with points:
The cost of a skill or spell, if you give its base and difficulty, such as "HT/A"	The cost of a skill or spell, if you give a number instead
Dodge, from Basic Speed	The skill level and damage of each attack
Thrust and swing damage, from ST 1 to 100	Damage above ST 100, with <code>thr:</code> and <code>sw:</code>

The point total	Discounts, such as the SM discount to ST
-----------------	--

A value that you give always replaces the calculated value. To give the cost of an attribute, give a dictionary: `st: (level: 20, points: 90)`. If an advantage or disadvantage has no points:, the stat block shows [?] and the total counts it as 0.

```
#import "@preview/gurps-ink:0.1.0": *

#stat-block(character(
  name: "The Tortoise",
  ht: 13,
  advantage("Fit", points: 5),
  advantage("Patience"),
  skill("Hiking", 14, "HT/A"),
  skill("Fast-Talk", 12, 4),
))
```

The Tortoise (43 points)
 ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 13 [30].
 HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 13 [0].
 Basic Speed 5.75 [0]; Basic Move 5 [0]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.
Advantages: Fit [5]; Patience [?].
Skills: Fast-Talk-12 [4]; Hiking-14 [4].

A skill can also have another skill of the same character as its base, such as "Running/E".

5.3 Hide point costs

Players do not need point costs in a handout or a bestiary. Use `show-points: false`.

```
#import "@preview/gurps-ink:0.1.0": *

#stat-block(
  character(
    name: "The Hare",
    dx: 13,
    skill("Running", 16, "HT/A"),
  ),
  show-points: false,
)
```

The Hare
 ST 10; DX 13; IQ 10; HT 10.
 HP 10; Will 10; Per 10; FP 10.
 Basic Speed 5.75; Basic Move 5; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.
Skills: Running-16.

5.4 Write a character's numbers in your text

Use `level-of(char, name)` for the level of an attribute, skill or spell. Use `char.thr` and `char.sw` for the damage. If you change the character, the text changes too.

```
#import "@preview/gurps-ink:0.1.0": *

#let hare = character(
  st: 6, dx: 13,
  basic-move: 9,
  skill("Running", 16, "HT/A"),
)

The Hare has Move
#level-of(hare, "Basic Move") and
Running-#level-of(hare, "Running").
It kicks for #dice(..hare.thr) cr.
```

The Hare has Move 9 and Running-16. It kicks for 1d-4 cr.

5.5 Use one character in many chapters

Put all characters in one dictionary. Then each chapter can use the same characters.

```
#import "@preview/gurps-ink:0.1.0": *

#let npcs = (
  tortoise: character(
    name: "The Tortoise", ht: 13),
  hare: character(
    name: "The Hare", dx: 13),
)

The Tortoise has
#level-of(npcs.tortoise, "HT") HT.
The Hare has
#level-of(npcs.hare, "DX") DX.
```

The Tortoise has 13 HT. The Hare has 13 DX.

Put the import line and the `#let npcs = (..)` lines in a file, for example `npcs.typ`. In each chapter, write `#import "npcs.typ": npcs.`

5.6 Keep a character in a point budget

Use `assert` with `total-points`. If the character costs too much, compilation stops and shows your message. This is useful for a pregenerated character or a template.

```
#import "@preview/gurps-ink:0.1.0": *

#let tortoise = character(
  name: "The Tortoise",
  iq: 11, ht: 13,
  basic-move: 1,
)
#assert(
  total-points(tortoise) <= 50,
  message: "Over 50 points",
)
#stat-block(tortoise)
```

The Tortoise (30 points)
 ST 10 [0]; DX 10 [0]; IQ 11 [20]; HT 13 [30].
 HP 10 [0]; Will 11 [0]; Per 11 [0]; FP 13 [0].
 Basic Speed 5.75 [0]; Basic Move 1 [-20]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.

5.7 Keep a stat block in one column

A stat block can break across columns and pages. To prevent this, put it in a block that cannot break.

```
#import "@preview/gurps-ink:0.1.0": *

#block(
  breakable: false,
  stat-block(character(
    name: "The Hare")),
)
```

The Hare (0 points)
 ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 10 [0].
 HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 10 [0].
 Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.

5.8 Change the look of a stat block

Use the hooks of `stat-block()`. The title hook makes the first line. The section hook makes each list, such as “Skills”. To frame a stat block, put it in a block with a `stroke`.

```
#import "@preview/gurps-ink:0.1.0": *

#block(stroke: 0.5pt, inset: 6pt,
  stat-block(
    character(
      name: "The Hare",
      skill("Running", 16, "HT/A"),
    ),
    title: (char, total) => smallcaps(
      [#char.name, #total points],
    ),
    section: (label, body) => [
      #emph(label) --- #body
    ],
  ),
))
```

THE HARE, 24 POINTS
 ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 10 [0].
 HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 10 [0].
 Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.
Skills — Running-16 [24].

5.9 Make a cast list or another layout

Write a function that takes the dictionary from `character()`. The reference for `character()` (Section 7.3) shows all keys of this dictionary.

```
#import "@preview/gurps-ink:0.1.0": *

#let cast(char) = {
  let skills = char.skills
  .map(s => [#s.name\-#s.level])
  [*#char.name* (#total-points(char)
    points): #skills.join[, ].]
}

#cast(character(
  name: "The Tortoise", ht: 13,
  skill("Hiking", 14, "HT/A"),
))

#cast(character(
  name: "The Hare", dx: 13,
  skill("Running", 16, "HT/A"),
))
```

The Tortoise (34 points): Hiking-14.
The Hare (84 points): Running-16.

When Typst gets user-defined elements, `stat-block` will become one. Then show and set rules can change it, as they change built-in elements.

5.10 Understand an error

When an input is wrong, compilation stops. The message tells you what is wrong. This table shows the frequent messages.

The message contains	Do this
is too low to buy	Make the skill level higher, or give the points as a number.
cannot work out the cost of	Make sure that the base ("HT" in "HT/A") is the name of an attribute, skill or spell of the same character.
more than one skill or spell is called	Give each skill and spell a different name.
got unknown argument(s)	Make sure that you spell the argument correctly.
covers ST 1 to 100	Give thr: and sw: to <code>character()</code> .

points must be zero or negative	Give a disadvantage a negative cost, for example points : -10.
---------------------------------	---

6 Example: The Tortoise and the Hare

This one-shot adventure uses each part of gurps-ink. Its source is [docs/example/tortoise-and-hare.typ](#). The PDF is [docs/tortoise-and-hare.pdf](#).

An example, not a tested adventure. This one-shot shows what the gurps-ink package can do: stat blocks, dice, book references and the Steve Jackson Games notices. Nobody has playtested it. Check the numbers before you use it at your table.

The Tortoise and the Hare

A one-shot adventure for *GURPS* Fourth Edition

One player and a GM · About 45 minutes · A 50-point hero

Introduction

Everybody knows how this race ends. Or do they? In this one-shot, the oldest race in the world is run again, and this time the dice decide. One player is the Tortoise: slow, stubborn and very hard to stop. The GM plays the Hare: the fastest animal in the meadow, and the first to say so.

The adventure is short. It fits in a lunch break. It needs one player, one GM, three six-sided dice and the *GURPS Basic Set*. New players learn Quick Contests and self-control rolls.

A Hare one day ridiculed the short feet and slow pace of the Tortoise, who replied, laughing: "Though you be swift as the wind, I will beat you in a race."
— Aesop, "The Hare and the Tortoise", translated by George Fyler Townsend (1867). Each quotation in this style is from this fable.

How to Use This Adventure

Read it once before you play. Give the player the Tortoise's stat block, or let the player build a Tortoise on 50 points. Then read "The Challenge" aloud, and play the race.

The Meadow

The meadow lies between the old oak and the mill pond. Rabbits, mice and a very old Fox live there. Nothing much happens in the meadow, and the animals like it that way. Then the Hare starts to boast.

The Challenge

Read aloud: "Under the old oak, the Hare stretches its long legs. 'Not one of you,' it says, 'could beat me from here to the mill pond.' The mice look away. The rabbits look at their feet. Then a slow voice comes from the grass: 'I could.'"

The Hare laughs so hard that it falls over. Then it accepts.

The Hare, believing her assertion to be simply impossible, assented to the proposal; and they agreed that the Fox should choose the course and fix the goal.
— Aesop

The Racers

The Tortoise

The Tortoise has a shell like a cobblestone and a will like a mule. It cannot run at all. It walks, it does not stop and it does not tire: that is its Hiking skill.

The Tortoise (50 points)

ST 7 [-30]; DX 10 [0]; IQ 11 [20]; HT 13 [30].

HP 10 [6]; Will 13 [10]; Per 11 [0]; FP 13 [0].

Basic Speed 5.75 [0]; Basic Move 1 [-20]; Dodge 8.

SM -3; DR 3 (shell); thr 1d-3; sw 1d-2.

Advantages: Damage Resistance (Tough Shell) 3 [15]; Very Fit [15].

Disadvantages: Stubbornness [-5].

Quirks: Never hurries [-1].

Skills: Fast-Talk-12 [4]; Hiking-14 [4]; Survival (Woodlands)-11 [2].

The Hare

The Hare is the fastest animal in the meadow. It also has Overconfidence (p. B148): it cannot believe that it could lose.

The Hare (59 points)

ST 6 [-40]; DX 13 [60]; IQ 10 [0]; HT 11 [10].

HP 6 [0]; Will 10 [0]; Per 10 [0]; FP 11 [0].

Basic Speed 6.00 [0]; Basic Move 9 [15]; Dodge 9.

SM -2; DR 0; thr 1d-4; sw 1d-3.

Disadvantages: Overconfidence (12) [-5].

Quirks: Naps after lunch [-1].

Skills: Running-16 [20].

Attacks:

Kick (13): 1d-4 cr. Reach C.

The Course

The Fox walks the course and plants a feather at the goal. It is one mile long and has three landmarks.

- **The clover patch.** Halfway. The clover is sweet and the sun is warm. It is a very good place for a nap.
- **The hedge.** A thick hawthorn hedge with one small gap at the bottom. At SM -3, the Tortoise fits through the gap. At SM -2, the Hare does not, and must run around.
- **The mill pond.** The goal. The Fox waits there to judge the finish.

The Race

The Tortoise never for a moment stopped, but went on with a slow but steady pace straight to the end of the course.
— Aesop

Play the race in three steps.

- 1. **Taunt (optional).** At the start, the Tortoise can mock the Hare. Roll a Quick Contest (p. B348) of the Tortoise's Fast-Talk-12 against the Hare's Will of 10. If the Tortoise wins, the Hare takes −4 on its roll in step 2.
- 2. **The nap.** At the clover patch, the Hare is far ahead. The GM rolls 3d against the Hare's self-control number of 12 (p. B120). On a failure, the Hare lies down for a nap.
- 3. **The finish.** Roll a Quick Contest: the Tortoise's Hiking-14 (p. B200) against the Hare's Running-16 (p. B218). The Hare gets +4 for its speed. If it took a nap, it gets −4 instead. If the player says that the Tortoise uses the gap in the hedge, the Tortoise gets +2. The winner reaches the pond first. On a tie, the Tortoise wins by the length of its nose.

Designer's note. These odds are calculated (see the appendix), not playtested. If the player only walks, the Tortoise wins about 1 race in 4. With the hedge, a little over 1 in 3. With the taunt, nearly 1 in 2. With the taunt and the hedge, about 6 in 10. Even against the hedge, an awake Hare wins about 4 races in 5, and a sleeping Hare loses about 6 in 7. Clever play pays, as it should.

The Hare, lying down by the wayside, fell fast asleep.
— Aesop

After the Race

If the Tortoise wins, the meadow cheers, and the Hare is quiet for a whole week. If the Hare wins, it boasts until the leaves fall. Either way, the story is over: this one-shot does not lead to a campaign.

Slow but steady wins the race.
— Aesop

Variations

- **More players.** Each extra player is another slow animal: a snail, a hedgehog or a mole. Each one rolls its own Quick Contest against the Hare.
- **A rematch.** The Hare has learned its lesson. It has no Overconfidence, so skip step 2. Can the Tortoise still win?

Appendix: The Odds

The designer's note quotes exact odds, not playtest results. This Python program calculates them from the rules of the *GURPS Basic Set*: success rolls, critical rolls and Quick Contests (p. B348). Change the numbers at the top to test your own version of the race.

The player	The Tortoise wins
Only walks	26%
Uses the gap in the hedge	37%
Taunts the Hare	45%
Taunts the Hare and uses the gap	57%

Exact odds for the race in The Tortoise and the Hare.

Rules (GURPS Basic Set): a success roll is 3d against the effective skill; 3 and 4 always succeed, 17 fails at skill 15 or less, 18 always fails (p. B348). In a Quick Contest both sides roll; the one who succeeds by more, or fails by less, wins (p. B348). A tie in the taunt has no effect; a tie at the finish goes to the Tortoise.

```
from itertools import product

# The numbers in the adventure.
FAST_TALK, HARE_WILL = 12, 10 # step 1: the taunt
TAUNT_PENALTY = 4 # to the Hare's self-control roll
SELF_CONTROL = 12 # step 2: Overconfidence (12)
HIKING, RUNNING = 14, 16 # step 3: the finish
AWAKE_BONUS, NAP_PENALTY = 4, 4 # the Hare's speed, or its nap
HEDGE_BONUS = 2 # the Tortoise uses the gap in the hedge

P3D = {}
for dice in product(range(1, 7), repeat=3):
    P3D[sum(dice)] = P3D.get(sum(dice), 0) + 1 / 216

def roll(skill, total):
    """(success, margin) of one success roll."""
    if total <= 4:
        success = True
    elif total == 18 or (total == 17 and skill <= 15):
        success = False
    else:
        success = total <= skill
    return success, skill - total

def p_success(skill):
    return sum(p for total, p in P3D.items() if roll(skill, total)[0])

def p_first_wins_or_ties(a, b):
    """Return (P(a wins), P(tie)) in a Quick Contest of a against b."""
    win = tie = 0.0
    for ta, pa in P3D.items():
        sa, ma = roll(a, ta)
        for tb, pb in P3D.items():
            sb, mb = roll(b, tb)
            if (sa and not sb) or (sa == sb and ma > mb):
                win += pa * pb
            elif sa == sb and ma == mb:
```

```
tie == pa * pb
return win, tie

def p_tortoise_wins(taunt, hedge):
    p_taunt = p_first_wins_or_ties(FAST_TALK, HARE_WILL)[0] if taunt else 0
    tortoise = HIKING + (HEDGE_BONUS if hedge else 0)
    total = 0.0
    for taunted, p1 in ((True, p_taunt), (False, 1 - p_taunt)):
        p_nap = 1 - p_success(SELF_CONTROL - (TAUNT_PENALTY if taunted else 0))
        for napped, p2 in ((True, p_nap), (False, 1 - p_nap)):
            hare = RUNNING + (-NAP_PENALTY if napped else AWAKE_BONUS)
            total += p1 * p2 * sum(p_first_wins_or_ties(tortoise, hare))
    return total

if __name__ == "__main__":
    for taunt, hedge, label in (
        (False, False, "only walks"),
        (False, True, "uses the hedge"),
        (True, False, "taunts"),
        (True, True, "taunts and uses the hedge"),
    ):
        print(f"Tortoise {label}: wins {p_tortoise_wins(taunt, hedge):.0%}")
    tortoise = HIKING + HEDGE_BONUS
    awake = 1 - sum(p_first_wins_or_ties(tortoise, RUNNING + AWAKE_BONUS))
    asleep = sum(p_first_wins_or_ties(tortoise, RUNNING - NAP_PENALTY))
    print(f"Against the hedge, an awake Hare wins {awake:.0%}")
    print(f"Against the hedge, a sleeping Hare loses {asleep:.0%}")
```

About This Adventure

Author. Nathanael Farley, October 2026.

Use of AI. Claude Opus 5.5, an AI model by Anthropic, was used to draft the text, the characters and the race rules, to calculate the odds, and to typeset the adventure. Nathanael Farley gave the directions and reviewed the result.

Tools. Typeset with Typst and gurps-link 0.1.0. The source of this adventure is docs/example/tortoise-and-hare.typ in the gurps-link repository. The appendix gives the program that calculates the odds in the designer's note.

References.

- Aesop, "The Hare and the Tortoise", in *Aesop's Fables*, translated by George Fyler Townsend (1867). Public domain. The italic quotations come from this translation.
- *GURPS Basic Set*, Fourth Edition, Steve Jackson Games. Rules cited: self-control rolls (p. B120), Overconfidence (p. B148), Hiking (p. B200), Running (p. B218), Quick Contests (p. B348).
- Steve Jackson Games, *Online Policy*: the disclaimer and the notice below.
- Steve Jackson Games, *Authors' Guidelines*: *Editorial Style*: dice, book titles and page references.

The material presented here is my original creation, intended for use with the *GURPS* system from Steve Jackson Games. This material is not official and is not endorsed by Steve Jackson Games.

GURPS is a registered trademark of Steve Jackson Games, and is copyrighted by Steve Jackson Games. All rights are reserved by Steve Jackson Games. This material is used here in accordance with the SJ Games online policy.

In the adventure	How it is made
Fonts, colours and page size of its own	Section 3.4
<i>GURPS</i> and <i>GURPS Basic Set</i> in the introduction	Section 3.3, Section 4.3
Two stat blocks that do not break	Section 5.1, Section 5.7
The Tortoise is built on 50 points	Section 5.6
Skill levels and SM in the text	Section 5.4
3d for the nap roll	Section 4.1
The disclaimer and the notice at the end	Section 3.1
Author, use of AI, tools and references at the end	Section 3.5
A box that says it is an untested example	Section 1

The source follows.

```
#import "@preview/gurps-ink:0.1.0": *

// Your own look: gurps-ink sets no fonts, colours or page layout.
#let accent = rgb("#2f5d50")
#set document(title: "The Tortoise and the Hare")
#set page(paper: "a5", margin: (x: 14mm, y: 15mm), numbering: "1")
#set text(size: 9pt)
#set par(justify: true)
#show heading: set text(fill: accent)
#show link: underline

// The old story, quoted beside the adventure. The text is the
// public-domain translation by George Fyler Townsend (1867).
#let aside(body, source: [Aesop]) = block(
  width: 100%,
  inset: (left: 8pt, y: 2pt),
  stroke: (left: 1.5pt + accent),
  text(fill: accent.darken(20%)) [
    #set par(justify: false)
    #emph(body)
    #linebreak()
    --- #source
  ],
)

// The two characters. Each section below uses them again.
#let tortoise = character(
  name: "The Tortoise",
  st: 7,
  iq: 11,
  ht: 13,
  hp: 10,
  will: 13,
  basic-move: 1,
  sm: -3,
  dr: [3 (shell)],
  advantage("Damage Resistance (Tough Shell)", points: 15, level: 3),
  advantage("Very Fit", points: 15),
  disadvantage("Stubbornness", points: -5),
  quirk("Never hurries"),
  skill("Hiking", 14, "HT/A"),
  skill("Fast-Talk", 12, "IQ/A"),
  skill("Survival (Woodlands)", 11, "Per/A"),
)

#let hare = character(
  name: "The Hare",
  st: 6,
  dx: 13,
  ht: 11,
  sm: -2,
  basic-move: 9,
  disadvantage("Overconfidence (12)", points: -5),
  quirk("Naps after lunch"),
```

```

skill("Running", 16, "HT/A"),
melee-attack("Kick", 13, [#dice(..thrust(6)) cr], reach: "C"),
)

// The player builds the Tortoise on 50 points. Compilation stops if a
// change makes it cost more.
#assert(total-points(tortoise) <= 50, message: "The Tortoise is over 50 points")

#block(
  fill: accent.lighten(88%),
  inset: 8pt,
  radius: 3pt,
  width: 100%,
  text(size: 8.5pt)[
    *An example, not a tested adventure.* This one-shot shows what the
    `gurps-ink` package can do: stat blocks, dice, book references and
    the #sjgames notices. Nobody has playtested it. Check the numbers
    before you use it at your table.
  ],
)

#align(center)[
  #text(size: 22pt, fill: accent, weight: "bold")[The Tortoise \ and the Hare]
  #v(2pt)
  #text(size: 11pt)[A one-shot adventure for #gurps Fourth Edition]
  #v(4pt)
  One player and a GM #sym.dot.c About 45 minutes #sym.dot.c A
  #total-points(tortoise)-point hero
]

#v(6pt)

```

== Introduction

Everybody knows how this race ends. Or do they? In this one-shot, the oldest race in the world is run again, and this time the dice decide. One player is the Tortoise: slow, stubborn and very hard to stop. The GM plays the Hare: the fastest animal in the meadow, and the first to say so.

The adventure is short. It fits in a lunch break. It needs one player, one GM, three six-sided dice and the #gurps-book("Basic Set"). New players learn Quick Contests and self-control rolls.

```

#aside(
  source: [Aesop, "The Hare and the Tortoise", translated by George Fyler
    Townsend (1867). Each quotation in this style is from this fable.],
)[
  A Hare one day ridiculed the short feet and slow pace of the Tortoise,
  who replied, laughing: "Though you be swift as the wind, I will beat you
  in a race."
]

```

== How to Use This Adventure

Read it once before you play. Give the player the Tortoise's stat block, or let the player build a Tortoise on #total-points(tortoise) points. Then read "The Challenge" aloud, and play the race.

== The Meadow

The meadow lies between the old oak and the mill pond. Rabbits, mice and a very old Fox live there. Nothing much happens in the meadow, and the animals like it that way. Then the Hare starts to boast.

== The Challenge

Read aloud: "Under the old oak, the Hare stretches its long legs. 'Not one of you,' it says, 'could beat me from here to the mill pond.' The mice look away. The rabbits look at their feet. Then a slow voice comes from the grass: 'I could.'"

The Hare laughs so hard that it falls over. Then it accepts.

```

#aside[

```

The Hare, believing her assertion to be simply impossible, assented to the proposal; and they agreed that the Fox should choose the course and fix the goal.

]

= The Racers

== The Tortoise

The Tortoise has a shell like a cobblestone and a will like a mule. It cannot run at all. It walks, it does not stop and it does not tire: that is its Hiking skill.

`#block(breakable: false, stat-block(tortoise))`

== The Hare

The Hare is the fastest animal in the meadow. It also has Overconfidence (p.~B148): it cannot believe that it could lose.

`#block(breakable: false, stat-block(hare))`

= The Course

The Fox walks the course and plants a feather at the goal. It is one mile long and has three landmarks.

- ***The clover patch.*** Halfway. The clover is sweet and the sun is warm. It is a very good place for a nap.
- ***The hedge.*** A thick hawthorn hedge with one small gap at the bottom. At SM `#level-of(tortoise, "SM")`, the Tortoise fits through the gap. At SM `#level-of(hare, "SM")`, the Hare does not, and must run around.
- ***The mill pond.*** The goal. The Fox waits there to judge the finish.

= The Race

`#aside[`

The Tortoise never for a moment stopped, but went on with a slow but steady pace straight to the end of the course.

`]`

Play the race in three steps.

- + ***Taunt (optional).*** At the start, the Tortoise can mock the Hare. Roll a Quick Contest (p.~B348) of the Tortoise's Fast-Talk-`#level-of(tortoise, "Fast-Talk")` against the Hare's Will of `#level-of(hare, "Will")`. If the Tortoise wins, the Hare takes -4 on its roll in step 2.
- + ***The nap.*** At the clover patch, the Hare is far ahead. The GM rolls `#dice(3)` against the Hare's self-control number of 12 (p.~B120). On a failure, the Hare lies down for a nap.
- + ***The finish.*** Roll a Quick Contest: the Tortoise's Hiking-`#level-of(tortoise, "Hiking")` (p.~B200) against the Hare's Running-`#level-of(hare, "Running")` (p.~B218). The Hare gets +4 for its speed. If it took a nap, it gets -4 instead. If the player says that the Tortoise uses the gap in the hedge, the Tortoise gets +2. The winner reaches the pond first. On a tie, the Tortoise wins by the length of its nose.

Designer's note. These odds are calculated (see the appendix), not playtested. If the player only walks, the Tortoise wins about 1 race in 4. With the hedge, a little over 1 in 3. With the taunt, nearly 1 in 2. With the taunt and the hedge, about 6 in 10. Even against the hedge, an awake Hare wins about 4 races in 5, and a sleeping Hare loses about 6 in 7. Clever play pays, as it should.

`#aside[`

The Hare, lying down by the wayside, fell fast asleep.

`]`

= After the Race

If the Tortoise wins, the meadow cheers, and the Hare is quiet for a whole week. If the Hare wins, it boasts until the leaves fall. Either

way, the story is over: this one-shot does not lead to a campaign.

`#aside[Slow but steady wins the race.]`

== Variations

- ***More players.*** Each extra player is another slow animal: a snail, a hedgehog or a mole. Each one rolls its own Quick Contest against the Hare.
- ***A rematch.*** The Hare has learned its lesson. It has no Overconfidence, so skip step 2. Can the Tortoise still win?

`#pagebreak()`

== Appendix: The Odds

The designer's note quotes exact odds, not playtest results. This Python program calculates them from the rules of the `#gurps-book("Basic Set")`: success rolls, critical rolls and Quick Contests (p.~B348). Change the numbers at the top to test your own version of the race.

```
#table(
    columns: (1fr, auto),
    stroke: none,
    inset: (x: 4pt, y: 3pt),
    table.hline(stroke: 0.4pt + accent),
    [*The player*], [*The Tortoise wins*],
    table.hline(stroke: 0.4pt + accent),
    [Only walks], [26%],
    [Uses the gap in the hedge], [37%],
    [Taunts the Hare], [45%],
    [Taunts the Hare and uses the gap], [57%],
    table.hline(stroke: 0.4pt + accent),
)

#[
    #set text(size: 7pt)
    #set par(justify: false)
    ```python
 """Exact odds for the race in The Tortoise and the Hare.

 Rules (GURPS Basic Set): a success roll is 3d against the effective
 skill; 3 and 4 always succeed, 17 fails at skill 15 or less, 18 always
 fails (p. B348). In a Quick Contest both sides roll; the one who succeeds
 by more, or fails by less, wins (p. B348). A tie in the taunt has no
 effect; a tie at the finish goes to the Tortoise.
 """

 from itertools import product

 # The numbers in the adventure.
 FAST_TALK, HARE_WILL = 12, 10 # step 1: the taunt
 TAUNT_PENALTY = 4 # to the Hare's self-control roll
 SELF_CONTROL = 12 # step 2: Overconfidence (12)
 HIKING, RUNNING = 14, 16 # step 3: the finish
 AWAKE_BONUS, NAP_PENALTY = 4, 4 # the Hare's speed, or its nap
 HEDGE_BONUS = 2 # the Tortoise uses the gap in the hedge

 P3D = {}
 for dice in product(range(1, 7), repeat=3):
 P3D[sum(dice)] = P3D.get(sum(dice), 0) + 1 / 216

 def roll(skill, total):
 """(success, margin) of one success roll."""
 if total <= 4:
 success = True
 elif total == 18 or (total == 17 and skill <= 15):
 success = False
 else:
 success = total <= skill
 return success, skill - total

 def p_success(skill):
```

```

 return sum(p for total, p in P3D.items() if roll(skill, total)[0])

def p_first_wins_or_ties(a, b):
 """Return (P(a wins), P(tie)) in a Quick Contest of a against b."""
 win = tie = 0.0
 for ta, pa in P3D.items():
 sa, ma = roll(a, ta)
 for tb, pb in P3D.items():
 sb, mb = roll(b, tb)
 if (sa and not sb) or (sa == sb and ma > mb):
 win += pa * pb
 elif sa == sb and ma == mb:
 tie += pa * pb
 return win, tie

def p_tortoise_wins(taunt, hedge):
 p_taunt = p_first_wins_or_ties(FAST_TALK, HARE_WILL)[0] if taunt else 0
 tortoise = HIKING + (HEDGE_BONUS if hedge else 0)
 total = 0.0
 for taunted, p1 in ((True, p_taunt), (False, 1 - p_taunt)):
 p_nap = 1 - p_success(SELF_CONTROL - (TAUNT_PENALTY if taunted else 0))
 for napped, p2 in ((True, p_nap), (False, 1 - p_nap)):
 hare = RUNNING + (-NAP_PENALTY if napped else AWAKE_BONUS)
 total += p1 * p2 * sum(p_first_wins_or_ties(tortoise, hare))
 return total

if __name__ == "__main__":
 for taunt, hedge, label in (
 (False, False, "only walks"),
 (False, True, "uses the hedge"),
 (True, False, "taunts"),
 (True, True, "taunts and uses the hedge"),
):
 print(f"Tortoise {label}: wins {p_tortoise_wins(taunt, hedge):.0%}")
 tortoise = HIKING + HEDGE_BONUS
 awake = 1 - sum(p_first_wins_or_ties(tortoise, RUNNING + AWAKE_BONUS))
 asleep = sum(p_first_wins_or_ties(tortoise, RUNNING - NAP_PENALTY))
 print(f"Against the hedge, an awake Hare wins {awake:.0%}")
 print(f"Against the hedge, a sleeping Hare loses {asleep:.0%}")
...
]

```

`#pagebreak()`

## **About This Adventure**

`#set text(size: 8pt)`

**\*Author.\*** Nathanael Farley, October 2026.

**\*Use of AI.\*** Claude Opus 5.5, an AI model by Anthropic, was used to draft the text, the characters and the race rules, to calculate the odds, and to typeset the adventure. Nathanael Farley gave the directions and reviewed the result.

**\*Tools.\*** Typeset with `#link("https://typst.app")` [Typst] and `#link("https://github.com/natfarleydev/gurps-typst")` [``gurps-ink``] 0.1.0. The source of this adventure is ``docs/example/tortoise-and-hare.typ`` in the ``gurps-ink`` repository. The appendix gives the program that calculates the odds in the designer's note.

## **\*References.\***

- Aesop, "The Hare and the Tortoise", in *Aesop's Fables*, translated by George Fyler Townsend (1867). Public domain. The italic quotations come from this translation.
- `#gurps-book("Basic Set")`, Fourth Edition, `#sjgames`. Rules cited: self-control rolls (p.~B120), Overconfidence (p.~B148), Hiking (p.~B200), Running (p.~B218), Quick Contests (p.~B348).
- `#sjgames`, `#link("https://www.sjgames.com/general/online_policy.html")` [Online Policy]: the disclaimer and the notice below.
- `#sjgames`,

```
#link("https://www.sjgames.com/general/guidelines/authors/style.html")[Authors'
 Guidelines: Editorial Style]: dice, book titles and page references.

#v(lfr)
#line(length: 100%, stroke: 0.4pt + accent)
#set text(size: 7pt)

#sjgames-disclaimer

#sjgames-notice
```

## 7 Reference

This chapter lists each function, in the same groups as chapters 3 to 5. Each example needs the import line first.

### 7.1 Legal notices and trademarks

#### 7.1.1 sjgames-game-aid

A notice for a free game aid that has no official license. The SJ Games online policy does not give this text. Many free GURPS game aids use it.

```
#sjgames-game-aid[Jane Doe]
```

**GURPS** is a trademark of [Steve Jackson Games](#), and its rules and art are copyrighted by [Steve Jackson Games](#). All rights are reserved by Steve Jackson Games. This game aid is the original creation of Jane Doe and is released for free distribution, and not for resale, under the permissions granted in the [Steve Jackson Games Online Policy](#).

#### Parameters

```
sjgames-game-aid(author: str content) -> content
```

**author** `str` or `content`

The author of the game aid.

#### 7.1.2 sjgames-disclaimer `content`

The disclaimer from the SJ Games online policy. It says that the material is yours and not official.

```
#sjgames-disclaimer
```

The material presented here is my original creation, intended for use with the **GURPS** system from [Steve Jackson Games](#). This material is not official and is not endorsed by Steve Jackson Games.

### 7.1.3 sjgames-notice content

The trademark notice from the SJ Games online policy. It does not mention art, because the package includes no SJ Games art.

#sjgames-notice

**GURPS** is a registered trademark of [Steve Jackson Games](#), and is copyrighted by Steve Jackson Games. All rights are reserved by Steve Jackson Games. This material is used here in accordance with the SJ Games [online policy](#).

### 7.1.4 gurps content

The name **GURPS** in bold italics. The SJ Games online policy asks for this style. Use it as you use other content.

#gurps is a generic, universal roleplaying system.

**GURPS** is a generic, universal roleplaying system.

To link it to the GURPS home page, put it in link: #link("http://www.sjgames.com/gurps/", gurps).

### 7.1.5 sjgames content

The name “Steve Jackson Games”.

#gurps is published by #sjgames.

**GURPS** is published by Steve Jackson Games.

## 7.2 Dice and book references

### 7.2.1 dice

Dice in the notation of the GURPS books. The result does not break across lines.

```
#dice(3) \
#dice(2, 1) \
#dice(4, -1) \
#dice(1, 0)
```

3d  
2d+1  
4d-1  
1d

The modifier is optional. A zero modifier shows nothing. A negative modifier has a true minus sign. A content modifier, such as \$m\$, gets a +.

To write basic damage, spread `thrust()` or `swing()` into it: `#dice(..thrust(13))`.

### Parameters

```
dice(
 count: int content,
 ..modifier: int content
) -> content
```

**count** int or content

The number of dice: an integer of 0 or more, or content such as \$n\$.

**..modifier** int or content

The modifier. It is optional; the default is 0.

### 7.2.2 thrust

The thrust damage for a ST, as a (count, modifier) pair. Spread it into `dice()`.

```
#thrust(10) \
#dice(..thrust(10))
```

```
(1, -2)
1d-2
```

It covers ST 1 to 100. Above ST 40, the Basic Set table has a row for each 5 ST; between two rows, the lower row applies. For a ST that is not in this range, compilation stops: give thr: to `character()`.

### Parameters

```
thrust(st: int) -> array
```

**st** int

The striking strength.

### 7.2.3 swing

The swing damage for a ST, as a (count, modifier) pair. Spread it into `dice()`. It covers the same ST as `thrust()`.

```
#dice(..swing(10))
```

```
1d
```

### Parameters

```
swing(st: int) -> array
```

**st**   **int**

The striking strength.

### 7.2.4 gurps-book

A reference to a GURPS book in the house style of SJ Games: the title in bold italics, then the pages if you give them.

```
#gurps-book("High Tech") \
#gurps-book("Zombies", 3) \
#gurps-book("Warehouse 23", "1, 3-5") \
#gurps-book("Basic Set", (16, 170))
```

***GURPS High Tech***  
***GURPS Zombies*** p. 3  
***GURPS Warehouse 23*** pp. 1, 3-5  
***GURPS Basic Set*** pp. 16, 170

One page (an int, or a string of digits) gets “p.”. All other pages get “pp.”. Commas join the pages in an array.

#### Parameters

```
gurps-book(
 title: str content,
 ..pages: int str content array
) -> content
```

**title**   **str** or **content**

The title of the book, without “GURPS”.

**..pages**   **int** or **str** or **content** or **array**

The page or pages. They are optional.

## 7.3 Characters and stat blocks

### 7.3.1 character

Makes a character. Give only the values that are not the default.

Give each attribute as a level. `character()` calculates its cost from the Basic Set rules. To give a different cost, such as an SM discount to ST, give a dictionary: `st: (level: 20, points: 90)`.

Give the traits, skills, spells and attacks as positional arguments.

```
#let napoleon = character(
 name: "Napoleon",
 st: 9, hp: 12,
 advantage("Natural afro", points: 1),
 quirk("Big teeth"),
 skill("Nunchuck", 16, "DX/E"),
)
#napoleon.attributes.HP \
#total-points(napoleon) points
```

(level: 12, points: 6)  
16 points

The result is a dictionary. Its keys are public API. Read them, or use `level-of()`, to quote numbers or to make your own layout. The keys are:

- kind: "character"; name.
- attributes: keyed ST, DX, IQ, HT, HP, Will, Per, FP, Basic Speed, Basic Move, in that order; each (level:, points:).
- dodge, sm, dr, thr, sw: as given or calculated.
- advantages, perks, disadvantages, quirks: arrays of (kind:, name:, points:, level:) in the order given.
- skills, spells: arrays of (kind:, name:, level:, points:, base:, difficulty:). If the skill has a base, points is calculated.
- attacks: arrays of (kind:, name:, level:, damage:, notes:) plus reach (melee) or range (ranged).

A value that is not known is none.

### Parameters

```
character(
 name: str content none ,
 st: int dictionary ,
 dx: int dictionary ,
 iq: int dictionary ,
 ht: int dictionary ,
 hp: auto int dictionary ,
 will: auto int dictionary ,
 per: auto int dictionary ,
 fp: auto int dictionary ,
 basic-speed: auto int float dictionary ,
 basic-move: auto int dictionary ,
 dodge: auto int ,
 sm: int ,
 dr: int str content ,
 thr: auto array content ,
 sw: auto array content ,
 ..traits: dictionary
) -> dictionary
```

**name** str or content or none

The name. The stat block shows it in the title.

Default: none

**basic-speed** auto or int or float or dictionary

A multiple of 0.25.

Default: auto

**dodge** auto or int

The default is [Basic Speed] + 3. It has no point cost.

Default: auto

**sm** `int`

The Size Modifier.

Default: `0`

**dr** `int` or `str` or `content`

The Damage Resistance, such as 2 or [3 (torso only)].

Default: `0`

**thr** `auto` or `array` or `content`

The thrust damage: a (count, modifier) pair or content. The default is `thrust()` of ST.

Default: `auto`

**sw** `auto` or `array` or `content`

The swing damage, as for thr. The default is `swing()` of ST.

Default: `auto`

**..traits** `dictionary`

Results of `advantage()`, `disadvantage()`, `perk()`, `quirk()`, `skill()`, `spell()`, `melee-attack()` and `ranged-attack()`, in any order.

### 7.3.2 stat-block

Typesets a character from `character()` as a stat block, in the style of the SJ Games books.

The lines are: the title (name and point total), ST/DX/IQ/HT, HP/Will/Per/FP, Basic Speed/Basic Move/Dodge and SM/DR/thr/sw. Then come the lists: advantages, perks, disadvantages, quirks, skills, spells and attacks. Each list is in alphabetical order. An empty list does not show.

```
#stat-block(character(
 name: "Napoleon",
 st: 9, hp: 12,
 advantage("Natural afro", points: 1),
 quirk("Big teeth"),
 skill("Nunchuck", 16, "DX/E"),
 melee-attack("Punch", 18, [#dice(5) cr],
reach: "C, 1",
 notes: [Believe it!]),
))
```

**Napoleon** (16 points)  
ST 9 [-10]; DX 10 [0]; IQ 10 [0]; HT 10 [0].  
HP 12 [6]; Will 10 [0]; Per 10 [0]; FP 10 [0].  
Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.  
SM 0; DR 0; thr 1d-2; sw 1d-1.  
**Advantages:** Natural afro [1].  
**Quirks:** Big teeth [-1].  
**Skills:** Nunchuck-16 [20].  
**Attacks:**  
**Punch (18):** 5d cr. Reach C, 1. Believe it!

To frame it, put it in a block: `#block(stroke: 0.5pt, inset: 8pt)[#stat-block(c)]`.

Two hooks change parts of the stat block:

```
#stat-block(
 character(name: "Guard", st: 11,
skill("Spear", 12, "DX/A")),
 title: (char, total) =>
smallcaps[#char.name, #total points],
 section: (label, body) => [#emph(label)
--- #body],
)
```

GUARD, 18 POINTS  
ST 11 [10]; DX 10 [0]; IQ 10 [0]; HT 10 [0].  
HP 11 [0]; Will 10 [0]; Per 10 [0]; FP 10 [0].  
Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.  
SM 0; DR 0; thr 1d-1; sw 1d+1.  
*Skills — Spear-12 [8].*

For a different layout, write your own function that uses the dictionary from `character()`. The manual shows how.

## Parameters

```
stat-block(
 char: dictionary,
 show-points: bool,
 title: auto none function,
 section: auto function
) -> content
```

**char** dictionary

A result of `character()`.

**show-points** `bool`

Show the point costs ([10]) and the total. Use `false` for a handout or a bestiary.

Default: `true`

**title** `auto` or `none` or `function`

Makes the first line. The call is `title(char, total)`. `total` is the point total, or `none` if `show-points` is `false`. Return content or a string. To remove the line, return `none` or give `title: none`. `auto` is this function:

```
(char, total) => {
 let points = if total != none [#total points]
 if char.name == none { points }
 else [#strong(char.name)#if points != none [(#points)]]
}
```

Default: `auto`

**section** `auto` or `function`

Makes each list: Advantages, Perks, Disadvantages, Quirks, Skills, Spells and Attacks. The call is `section(label, body)`. `body` is the finished list: the entries, with semicolons between them and a full stop at the end. For attacks, `body` is a block with one paragraph for each attack. Return content or a string. To remove the list, return `none`. `auto` is `(label, body) => [#strong[#label:] #body]`.

Default: `auto`

### 7.3.3 advantage

An advantage.

```
#let afro = advantage("Natural afro",
points: 1)
#let magery = advantage("Magery", points:
25, level: 2)
#afro.points, #magery.level
```

1, 2

#### Parameters

```
advantage(
 name: str content,
 points: int none,
 level: int none
) -> dictionary
```

**points** `int` or `none`

The point cost. If it is `none`, the stat block shows [?] and the total counts it as 0.

Default: `none`

**level**   `int` or `none`

The level of an advantage that has levels, such as Magery 2.

Default: `none`

#### 7.3.4 disadvantage

A disadvantage. If you give a point cost, it must be zero or negative.

```
#disadvantage("Bad Temper", points:
-10).points
```

-10

##### Parameters

```
disadvantage(
 name: str content,
 points: int none,
 level: int none
) -> dictionary
```

**points**   `int` or `none`

The point cost, zero or less. If it is none, the stat block shows [?] and the total counts it as 0.

Default: `none`

**level**   `int` or `none`

The level of a disadvantage that has levels.

Default: `none`

#### 7.3.5 perk

A perk. Its cost is always 1 point.

##### Parameters

```
perk(name: str content) -> dictionary
```

#### 7.3.6 quirk

A quirk. Its cost is always -1 point.

##### Parameters

```
quirk(name: str content) -> dictionary
```

### 7.3.7 skill

A skill at a given level.

The third argument is the cost. It is optional. Give it in one of two forms:

- The points, such as 4.
- The base and the difficulty, such as "DX/E" or "Per/Average". Then `character()` calculates the points.

The base can be an attribute, or a skill or spell of the same character ("Typst/H"). The match uses the plain text of skill and spell names, so `[_Typst_]` matches "Typst". If a skill is a base for other skills, give it a name that has plain text.

The difficulty is E, A, H, VH or W, or the full word (Easy, Average, Hard, Very Hard, Wildcard).

```
#skill("Stealth", 12, "DX/A") \
#skill("Brawling", 14, 4) \
#skill("Carousing", 10)
```

```
(
 kind: "skill",
 name: "Stealth",
 level: 12,
 points: none,
 base: "DX",
 difficulty: "A",
)
(
 kind: "skill",
 name: "Brawling",
 level: 14,
 points: 4,
 base: none,
 difficulty: none,
)
(
 kind: "skill",
 name: "Carousing",
 level: 10,
 points: none,
 base: none,
 difficulty: none,
)
```

#### Parameters

```
skill(
 name: str content,
 level: int,
 ..cost: int str
) -> dictionary
```

**level** int

The skill level.

**..cost** `int` or `str`

The cost: points, or "<base>/<difficulty>". It is optional.

### 7.3.8 spell

A spell. It has the same arguments as `skill()`.

```
#spell("Fireball", 14, "IQ/H")
```

```
(
 kind: "spell",
 name: "Fireball",
 level: 14,
 points: none,
 base: "IQ",
 difficulty: "H",
)
```

#### Parameters

```
spell(
 name: str content,
 level: int,
 ..cost: int str
) -> dictionary
```

### 7.3.9 melee-attack

A melee attack.

```
#melee-attack("Punch", 18, [#dice(1, -1)
cr], reach: "C")
```

```
(
 kind: "melee-attack",
 name: "Punch",
 level: 18,
 damage: sequence(box(body: sequence([1,
[d], [-1]]), [], [cr])),
 notes: none,
 reach: "C",
)
```

#### Parameters

```
melee-attack(
 name: str content,
 level: int,
 damage: str content,
 reach: str content none,
 notes: str content none
) -> dictionary
```

**level** `int`

The skill level with the attack.

**damage** `str` or `content`

The damage, such as `[#dice(2, 1) cut]`.

**notes** `str` or `content` or `none`

Notes. The stat block shows them after the attack.

Default: `none`

### 7.3.10 ranged-attack

A ranged attack.

```
#ranged-attack("Bow", 14, [#dice(1, 2) imp],
range: "150/200")
```

```
(
 kind: "ranged-attack",
 name: "Bow",
 level: 14,
 damage: sequence(box(body: sequence([1,
[d], [+2]]), [], [imp])),
 notes: none,
 range: "150/200",
)
```

#### Parameters

```
ranged-attack(
 name: str content,
 level: int,
 damage: str content,
 range: str content none,
 notes: str content none
) -> dictionary
```

### 7.3.11 level-of

The level of an attribute, skill or spell, found by its name. For a skill or spell, it uses the plain text of the name. It also knows "Dodge" and "SM".

Compilation stops if no name matches, or if two names match. The message lists the names that the character has.

```
#let c = character(dx: 12, skill("Stealth",
13, "DX/A"))
#level-of(c, "Basic Speed"), #level-of(c,
"Stealth")
```

5.5, 13

### Parameters

```
level-of(
 char: dictionary,
 name: str
) -> int float
```

**char** dictionary

A result of `character()`.

**name** str

The name, such as "ST", "Basic Move" or "Stealth".

### 7.3.12 total-points

The point total of a character: the sum of all known costs. A trait that has no cost counts as 0.

```
#total-points(character(st: 12, dx: 11))
```

40

### Parameters

```
total-points(char: dictionary) -> int
```

**char** dictionary

A result of `character()`.

## 8 Legal

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